Math 520

Week 5 Homework

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Exercise 1. *Define* $T : c_0 \to c_0$ *by* $T(\{x_n\}_{n=1}^{\infty}) = \{x_{n+1}\}_{n=1}^{\infty}$. *Determine if* T *is: injective, surjective, open, closed, invertible*

Proof.

- Let $(x_1, x_2, x_3, ...)$ and $(y_1, y_2, y_3, ...)$ be sequences in c_0 and $\alpha \in \mathbb{F}$. Observe that:

$$\begin{aligned} T((x_1, x_2, x_3, ...) + \alpha(y_1, y_2, y_3, ...)) &= T((x_1 + \alpha y_1, x_2 + \alpha y_2, x_3 + \alpha y_3, ...)) \\ &= (x_2 + \alpha y_2, x_3 + \alpha y_3, x_4 + \alpha y_4, ...) \\ &= (x_2, x_3, x_4, ...) + \alpha(y_2, y_3, y_4, ...) \\ &= T((x_1, x_2, x_3, ...)) + \alpha T((y_1, y_2, y_3, ...)) \end{aligned}$$

  Hence T is linear.
- Consider the sequences $(1, 0, 0, ...)$ and $(2, 0, 0, ...)$. These are elements of c_0 which are not equal, but $T((1, 0, 0, ...)) = (0, 0, 0, ...) = T((2, 0, 0, ...))$. Therefore T is not injective.
- Let $(x_1, x_2, x_3, ...) \in c_0$. Then $T((0, x_1, x_2, x_3, ...)) = (x_1, x_2, x_3, ...)$. Thus T is surjective.
- Let $(x_1, x_2, x_3, ...) \in c_0$. Observe that:

$$\begin{aligned} \|T((x_1, x_2, x_3, ...))\|_{\ell^\infty} &= \|(x_2, x_3, x_4, ...)\|_{\ell^\infty} \\ &= \sup\{|x_i| \mid i = 2, 3, 4, ...\} \\ &\leqslant \sup\{|x_i| \mid i = 1, 2, 3, ...\} \\ &= \|(x_1, x_2, x_3, ...)\|_{\ell^\infty}. \end{aligned}$$

  Since T is bounded, T is continuous.
- Since T is surjective, continuous, and linear, by the Open Mapping Theorem T is open.
- Since T is continuous, T has a closed graph.
- Since T is not bijective, T is not invertible.

□

Exercise 2. *Let X be a normed linear space, let $K \subset X$, and let $\Omega \subset X^*$. Show that $K^\perp$ is closed in X^* and that $\Omega_\perp$ is closed in X.*

Proof. Let $\{\varphi_n\}_{n=1}^{\infty} \subset K^\perp$ be a sequence such that $\varphi_n \to \Phi \in X^*$. Let $k \in K$ be arbitrary. Then $\varphi_n(k) \to \Phi(k)$. Since $\varphi_n(k) = 0$ for all $n \in \mathbb{N}$, it must be the case that $\Phi(k) = 0$. Since k was arbitrary, we have $\Phi \in K^\perp$. Since $K^\perp$ contains all of its limit points, it is closed in X^*.

Let $\{x_n\}_{n=1}^{\infty} \subset \Omega_\perp$ be a sequence such that $x_n \to y \in X$. Let $\varphi \in \Omega$ be arbitrary. Then $\varphi(x_n) \to \varphi(y)$ since φ is continuous. Since $\varphi(x_n) = 0$ for all $n \in \mathbb{N}$, it must be the case that $\varphi(y) = 0$. Since φ was arbitrary, $y \in \Omega_\perp$. Since $\Omega_\perp$ contains all of its limit points, it is closed in X. □

Exercise 3. *For which $r \in \mathbb{R}$ is the function $f(x) = \frac{1}{|x|^r}$ in $L^1([0,1])$? What about $L^p([0,1])$?*

Proof. Since $x \in [0,1]$, we have $|x| = x$. Note that:

$$\int_0^1 x^{-r}\,dx = \left.\frac{x^{-r+1}}{-r+1}\right|_0^1$$

The right-hand side of this equation can only be evaluated if $-r+1 > 0$. Hence $\frac{1}{|x|^r} \in L^1([0,1])$ provided $r < 1$. Similarly, note that:

$$\begin{aligned}\int_0^1 (x^{-r})^p\,dx &= \int_0^1 x^{-rp}\,dx \\ &= \left.\frac{x^{-rp+1}}{-rp+1}\right|_0^1\end{aligned}$$

The right-hand side of this equation can only be evaluation if $-rp+1 > 0$. Hence $\frac{1}{|x|^r} \in L^p([0,1])$ provided $r < \frac{1}{p}$. □

Exercise 4. *Prove the Parallelogram Law in any inner-product space.*

Proof. Observe that:

$$\begin{aligned}\|x+y\|^2 + \|x-y\|^2 &= (x+y, x+y) + (x-y, x-y) \\ &= (x,x) + 2\Re((x,y)) + (y,y) + (x,x) - 2\Re((x,y)) + (y,y) \\ &= 2(x,x) + 2(y,y) \\ &= 2\|x\|^2 + 2\|y\|^2.\end{aligned}$$

□

Exercise 5. *Let X be an inner-product space over $\mathbb{C}$ and fix $y \in X$. Prove that the map $\varphi_y : X \to \mathbb{C}$ defined by $\varphi_y(x) = (x, y)$ is continuous.*

Proof. Let $x_1, x_2 \in X$. We have:

$$\begin{aligned}|\varphi_y(x_1) - \varphi_y(x_2)| &= |(x_1, y) - (x_2, y)| \\ &= |(x_1 - x_2, y)| \\ &\leqslant \|x_1 - x_2\|\,\|y\|.\end{aligned}$$

Since φ_y is Lipschitz, φ_y is continuous. □

Exercise 6. *Prove that in an inner-product space* X, *if* $\{x_n\}_{n=1}^{\infty} \subset X$ *is a sequence and* $y \in X$ *satisfy:*

$$\|x_n\| \to \|y\| \text{ and } (x_n, y) \to \|y\|^2,$$

then $x_n \to y$.

Proof. Observe that:

$$\begin{aligned}
\lim_{n\to\infty} \|x_n - y\|^2 &= \lim_{n\to\infty} (x_n - y, x_n - y) \\
&= \lim_{n\to\infty} \Big((x_n, x_n) - (y, x_n) - (x_n, y) + (y, y) \Big) \\
&= \lim_{n\to\infty} \Big(\|x_n\|^2 - 2\Re((x_n, y)) + \|y\|^2 \Big) \\
&= \lim_{n\to\infty} \|x_n\|^2 - \lim_{n\to\infty} 2\Re((x_n, y)) + \lim_{n\to\infty} \|y\|^2 \\
&= \lim_{n\to\infty} \|x_n\|^2 - 2\Re(\lim_{n\to\infty} (x_n, y)) + \lim_{n\to\infty} \|y\|^2 \\
&= \|y\|^2 - 2\|y\|^2 + \|y\|^2 \\
&= 0.
\end{aligned}$$

Since $\|x_n - y\|^2 \to 0$, it must be the case that $\|x_n - y\| \to 0$. Thus $x_n \to y$. □